Manual

Maintenance Calendar for Machines (MOC)

MDE-MWK 8.1

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Contents

1 Overview of Machine Maintenance Calendar

Purpose

The maintenance calendar is intended for planning and visualizing maintenance work. This overview is a valuable tool for foremen and maintenance personnel, because only regular maintenance will allow expensive resources (e.g. production machines) to continue to produce quality results and not cause any unnecessary downtimes.

Implementation considerations

You use the function package if you want to monitor machine maintenance intervals:

- based on cycles recorded
- based on durations recorded
- based on the Gregorian calendar

in order to subsequently initiate the relevant maintenance activities.

Integration

Under the MDE MWK license only machine maintenance work is permitted. Functions that relate to other resources, however, are not available there, or only to a limited extent. The latter requires the WRM WWR license.

Features

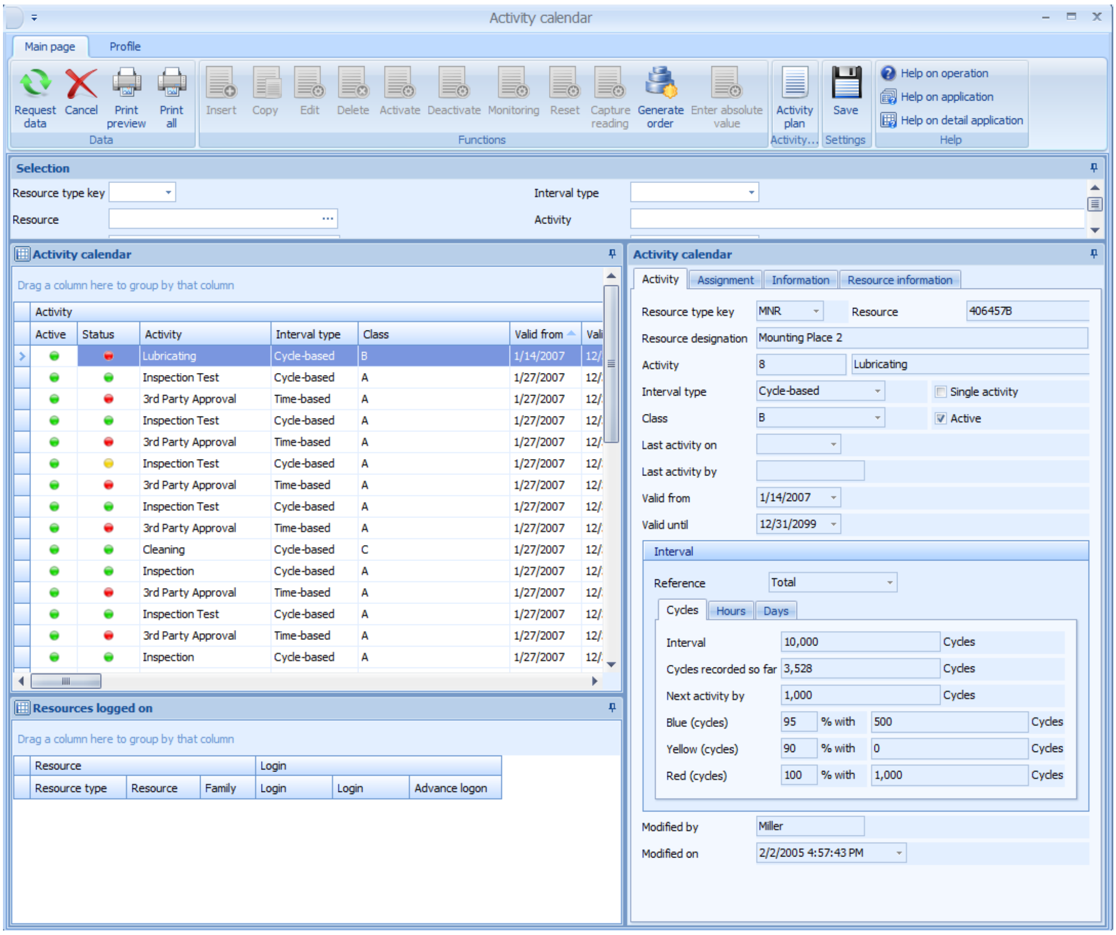
- Function for setting up a maintenance calendar allowing maintenance activities to be defined arbitrarily for each machine
- Definition of maintenance activities with a free choice of maintenance intervals with a color-coded breakdown (3 levels)
- Monitoring of maintenance activities by cycles, hours of operation (with the ability to configure the relevant resource performance accounts per resource type) and number of days
- Comparison of target and actual values for each maintenance
- Display in machine history of maintenance due dates and maintenance cycle violations.
- Display of a total maintenance status for assigned machines at Windows-based shop floor terminals
- Function at the HYDRA client to manually reset the actual values after completion of maintenance. The maintenance messages go into the machine history.

2 Maintenance Calendar - Activities Calendar

Summary

Menu	Production Facilities Management → Current Information → Maintenance Calendar
Transaction code	rmcal
Function authorization	rmcal

Only through regular maintenance work can expensive resources (e.g. production machines or tools) maintain their production quality and unnecessary downtimes be avoided; this overview, therefore, represents a valuable aid for the foreman and for maintenance personnel. In addition, the calendar also represents the basis for maintenance planning and at the same time has been designed as data acquisition tool to record the performed activities.



Usage

The maintenance calendar or activity calendar is used for the planning and visualization of maintenance work or similar, recurring activities. This includes, e.g. maintenance activities, gage calibrations, meter readings of manual energy counters, or similar. The "activity type" field is provided to distinguish the different functions of activity types. In most cases, nothing is entered here, which suggests maintenance or an activity relating to a similar resource. The relevant ID is entered here (e.g. ID "K" for gage calibration) for special requirements such as gage calibration. Consequently, activities can also be distinguished by their type.

This document describes the mode of operation of the calendar within the framework of the HYDRA Tool and Resource Management, Energy Management and Gage Calibration.

Monitoring of maintenances or activities is responsible for monitoring the configured activities and performing the below-mentioned actions:

Refreshing the current values
- "Previously recorded cycles" (cycle-oriented maintenance/activity) or
- "Previously recorded hours" (maintenance/activity of hours of operation).

Setting the status of an activity, when a set threshold value (blue/yellow/red) has been exceeded, and documentation of this event in the database (for purposes of evaluation via the resource history). Checking for exceeded threshold values follows the order: red > yellow >blue. This means, that a check is first made of whether the threshold "red" has been exceeded. If that is so, this status is set and documented. Otherwise, the inspection is continued for the threshold values "yellow" and then "blue".

Monitoring is only run for activities that are marked active and whose validity period includes the current time.

For this purpose, the monitoring process hywtkupd.out/.exe is embedded in the HYDRA Scheduler and run in cycles. .

Any number of activities can be defined for each resource. Several activities can thereby be defined for each resource. The following types of intervals are available when defining the maintenance times:

Cycle-oriented activity

With cycle-oriented activity, target and actual cycle times are compared. The size of the difference indicates when maintenance time has been reached. Actual cycle times are automatically recorded in HYDRA.

Cyclical monitoring of the machine is a precondition of this process

Activity based on hours of operation

The times recorded in HYDRA are used for maintenance/activity based on hours of operation. For this reason, it is determined in the resource type which resource performance accounts should be used for the calculation of the hours of operation. Activity is then due when the interval defined in the maintenance calendar has been reached.

Time-oriented activity (days)

With this type of activity, the next maintenance date is calculated on the basis of the number of days which is set as part of the definition of the activity. This number of days is based on the Gregorian calendar.

Non-recurring activity

Combined with the above-mentioned types of intervals, it is even possible to define an activity as "non-recurring". In this case, the activity is deactivated automatically, once it has been reset.

Please observe the following additional instructions

When data is selected, users can only view the activities of such resources for which they are authorized by the responsibility area.

The resource master data which is required for the maintenance calendar must be defined in the resource stock.

For the use of cycle-oriented activities and maintenance based on hours of operation, resources must be configured as available for assignment. This can be configured via the resource type.

For resources with DNC processing, no cycle-oriented activities and no activities based on hours of operation can be defined, as these resources cannot be assigned/posted.

This is neither possible for energy counters, as energy counters do not use machine cycles as the data basis.

Selection criteria

The application provides the following selection criteria:

Resource type

Selects the specified resource type. This field is pre-assigned to the value from the selected line, if the function is started from the resource stock.

Resource

Selects the specified resource. This field is pre-assigned to the value from the selected line, if the function is started from the resource stock.

Field description of the "activity" tab

Resource type

Resource type of the resource for which the activity is defined.

For the use of cycle-oriented activities and maintenance based on hours of operation, resources must be configured as available for assignment. This can be configured via the resource type.

For resource types with DNC processing, no cycle-oriented maintenance and no maintenance based on hours of operation can be defined, as these resources cannot be assigned (data cannot be posted onto them).

Resource

Resource for which the activity is defined.

Activity

Designation of the activity. The number in front of the name is automatically allocated when an activity is created and gives it a unique identity.

Type

Selection of a type determines how monitoring is to be carried out:

T Cycle-oriented activity
B Activity based on hours of operation
Z Time-oriented activity

Depending on this selection, one of the following tabs is enabled: Cycles, Hours of operation or Days.

Non-recurring activity

If this option is checked the activity is a non-recurring activity. In this case, the "interval" field is hidden. When the option is reset, the activity is automatically deactivated.

Class

This input field allows for maintenance activities to be classified. For example, all cleaning activities can be marked as "Cleaning".

Using the grouping function in the overview screen, all maintenance activities of the same class can be logically gathered together and displayed. Otherwise, this entry field is used for comments.

Active

It is possible to deactivate activities temporarily, and reactivate it again at a later time. The activation state of an activity is indicated by this display:

 Activity enabled
 Activity disabled

Disabled activities are not taken into consideration during monitoring.

Status

A graphic display, in the form of a lamp, is placed before every activity, and clearly indicates if a maintenance interval will soon expire or has already expired, and that a maintenance activity must therefore be carried out. This allows the user to recognize quickly which activity is to be carried out or is already overdue.

For this purpose, threshold values, which cause the color of the lamp to change, are set (as a percentage for cycle-oriented activities and maintenance based on hours of operation and as the number of days for time-oriented maintenance). The following 4 colors can be defined (each corresponding to a status type):

green
blue
yellow
red

Additional information on this topic is located in the descriptions of the individual interval types (see below).

Please note: The status of the resource itself does not change when a maintenance status is reached.

Last activity

This shows the time at which the maintenance activity was last reset and the name of the user who reset the maintenance.

Please note that the time of resetting the maintenance may deviate from the time that the maintenance is actually carried out.

Valid from, valid to

Maintenance activities can be assigned time limits by setting these values.

Maintenance activities whose validity period does not include the current time are not taken into consideration during maintenance monitoring, i.e. their maintenance status will not be updated.

Editor

Name of the last user to edit the maintenance activity as well as the time of the last change.

The individual tabs are explained below.

Field description of the "cycles" sub-tab

Interval

The number of machine cycles after which maintenance is to be carried out is to be entered here. This number, and the following two numbers, refer to the value in the Reference field (see below).

This field is hidden if it is a one-off/non-recurring maintenance.

Previously recorded cycles

The number of machine cycles recorded so far in HYDRA is displayed here. This value is updated by a cyclical process. Additional information on this is contained in the section *Maintenance monitoring*.

Next activity after

When a new activity is created, this value is calculated by default from the current actual value (actual cycles) + the specified interval.

When the maintenance activity is reset, this value is calculated and points out when the next activity is due for this maintenance activity.

When an activity is reset, the "calculation base" option can be used to determine how the next maintenance due date is to be calculated:

Target value	The value for the next activity is based on the current value: Next activity after = current value (next activity after) + interval
Actual value	The value for the next activity is based on the number of previously recorded cycles: Next activity after = previously recorded cycles + interval

Reference

This option must generally be taken into account for the previous values. The following values are possible here:

G	Total Activity monitoring is based on the total number of cycles previously recorded.
A	Relating to order/OP If a resource is logged on by means of an operation logon, then a check is made to ensure that activities with reference = A exist for this resource. These activities are then automatically reset. On the basis of cycles posted for the currently logged on operation, "monitoring" now checks whether the interval has been reached, and then sets the status accordingly.

Order related monitoring is not available for resources of the type "MNR“ (machines)!!

This type of maintenance monitoring only makes sense for resources that are logged on to no more than one operation at any given time. This means that a maximum of one operation may be logged on to the workstation/machine.

The posting of cycles to a resource does not take place in real time, but at longer intervals (e.g. at logoff, at interruption of an operation or during an automatic change of shifts). This type of monitoring is, therefore, only meaningful for operations that have longer runtimes.

Blue / Yellow / Red

The threshold values, which determine the status of a maintenance activity, can be entered here as percentages.

"Previously recorded cycles" < Blue-% from "Next maintenance after"	green
"Previously recorded cycles" >= Blue-% and < Yellow-% from "Next maintenance after"	blue
"Previously recorded cycles" >= Yellow-% and < Red-% from "Next maintenance after"	yellow
"Previously recorded cycles" >= Red-% from "Next maintenance after"	red

The color of the graphic display, or "lamp", depends on the values entered.

Please note

The threshold values can be greater than 100%.

No validation check is made with regard to the order of the threshold values.

Field description of the "operating hours" sub-tab

Interval

The period of time, after which the maintenance activity is to be run, should be entered here. This value, and the two following values, refer to the value in the Reference field (see below).

This field is hidden if it is a one-off/non-recurring activity.

Previously recorded hours

The time previously posted in HYDRA for this resource is shown here. This value is updated by a cyclical process.

It is to be observed here that, for the previously recorded hours, only those RPA times are used, which have been marked as such in the resource type (option: *RPAs as hours of operation in the Maintenance Calendar*).

Next activity after

On the creation of a new maintenance activity, this value is calculated by default from the current actual value (previously recorded hours) + the specified interval.

When the activity is reset, this value is calculated and points out when the next activity is due for this maintenance activity.

When a maintenance activity is reset, the "calculation base" option can be used to determine how the next activity is to be calculated:

Target value	The value for the next activity is based on the current value: Next activity after = current value (next activity after) + interval
Actual value	The value for the next activity is based on the number of previously recorded hours: Next activity after = previously recorded hours + interval

Reference

This option must generally be taken into account for the previous values. The following values are possible here:

G Total
Activity monitoring is based on the total time that has been posted so far.

A Relating to order/OP
If a resource is logged on by means of an operation logon, then a check is made to ensure that activities with reference = A exist for this resource. These activities are then automatically reset.

On the basis of the duration posted for the currently logged on operation, monitoring now checks whether the interval has been reached, and then sets the status accordingly.

This type of maintenance monitoring only makes sense for resources that are logged on to no more than one operation at any given time. This means that a maximum of one operation may be logged on to the workstation/machine.

The posting of cycles to a resource does not take place in real time, but at longer intervals (e.g. at logoff, at interruption of an operation or during an automatic change of shift). This type of monitoring is, therefore, only meaningful for operations that have longer runtimes.

Blue / Yellow / Red

The threshold values, which determine the status of a maintenance activity, can be entered here as percentages.

"Previously recorded hours" < Blue-% from "Next activity after"	green
"Previously recorded hours" >= Blue-% and < Yellow-% from "Next activity after"	blue
"Previously recorded hours" >= Yellow-% and < Red-% from "Next activity after"	yellow
"Previously recorded hours" >= Red-% from "Next activity after"	red

The color of the graphic display, or "lamp", depends on the values entered.

Please note

The threshold values can be greater than 100%. No validation check is made with regard to the order of the threshold values.

Field description of the "days" sub-tab

Interval

Interval in days after which an activity is to be run. This interval is based on the Gregorian calendar.

This field is hidden if it is a one-off/non-recurring activity.

Next activity after

Date when the next activity falls due.

When a new maintenance is created, this value is calculated by default from the current date + the specified interval.

Blue / Yellow / Red

The number of days, which determines the status of a maintenance activity, can be entered here. The color of the lamp is based on the remaining time, i.e. the difference between the date of the next activity and the current date ("today").

Remaining time <= "Red" value	red
Remaining time <= "Yellow" value	yellow
Remaining time <= "Blue" value	blue
Other	green

Please note

The threshold values can be greater than 100%.

No validation check is made with regard to the order of the threshold values.

Field description of the "assignment" tab

Order

This field is only relevant in connection with the additional feature "generate maintenance orders" or the "generation of calibration (inspection) orders. A maintenance order/calibration order is assigned by using the "create order" function that can be started using the button .

If this field is filled out the included order number refers to a maintenance/calibration order. The activity will automatically be reset if the maintenance/calibration order is finished. As the maintenance/calibration order is finished for this activity, the order number is also removed from this input field.

Project number

This field is only relevant to the activity type "K" (calibration), whereas there are two different variants subject to system configuration.

Variant 1 (there is exactly one work plan for all calibration inspection plans):

=> Input of the calibration inspection plan number (without taking the version number into account)

Variant 2 (there is a separate work plan for each calibration inspection plan)

=> Should be left empty. If it is filled out, this project number will be assigned by default for the work plan list to be opened within the "order generation" application.

Planned order

Control field that is currently not used. Consequently it remains empty.

Cost object

Control field that is currently not used. Consequently it remains empty.

Activity type

Identifies the activity type, calibrations, for example, are identified by the type K.

Description of the "information" tab

To ensure that the user or the maintenance worker receives more detailed information about running the activity (e.g. notes on regulations to be observed, materials to be used), a short description can be stored for each maintenance activity.

Toolbar

Activate

Function authorization: rmcal.active

Opens the editing dialog to enable a disabled activity

Deactivate

Function authorization: rmcal.deactive

Opens the editing dialog disable an enabled activity

Monitoring

Function authorization: rmcal.monitor

Updates the activity statuses

Reset

Function authorization: rmcal.reset

Opens the editing dialog to reset an activity

Capture reading (EMG 8.1 only)

Function authorization: rmcal.captvalues

Opens the editing dialog to enter the meter reading. Any required number of difference values may be entered within a time interval. Consequently, it is about delta collection. For this reason, it is also possible to subsequently enter data relating to the past. If the date fields are not entered, the system independently generates the interval from the end of the last data capture as the start time and the current posting time as the end of the interval.

Enter absolute values (EMG 8.1 only)

Function authorization: rmcal.captvalues

Opens the editing dialog to enter absolute meter values. The system calculates the difference to the previous meter reading. It is not possible to enter values for past periods. The system sets the start time of the interval to the end of the last data capture. The system uses the current posting time if no end is entered.

Generate order

Function authorization: rmcal.generate

Generates an order by which the activity is to be processed from an organizational point of view. Once the order is finished, the activity can be reset automatically.

Activity plan (EMG 8.1 only)

Function authorization: rmcal.timetable

Opens the report activity plan

"Resources logged on" detail application

The detail application provides additional information about the resources logged on to the machine for resources of the type "machine".

Resource type, resource

The currently selected resource and its resource type.

Resource, family, resource type

Resource that is logged on and the family as well as the resource type.

Logon

Date and time of the resource login

Advance logon

Advance logon flag